UNIVERSITY OF SCIENCE AND TECHNOLOGY OF HANOI

COMPUTER VISION PROJECT REPORT

Receipts OCR and Information Extraction

Group 6

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List of Abbreviations

LLM	Large Language Model
AI	Artificial Intelligence
ML	Machine Learning
MLP	Multilayer Perceptron

Abstract

This project studies receipt understanding on the SROIE task. The goal is to extract four key fields from receipt images, namely company, date, address, and total. Instead of relying on a single approach, the current implementation explores three directions. The first is a rule-based baseline applied at test time. The second is a layout-aware extraction branch based on LayoutLMv3. The third is an end-to-end document understanding branch based on DONUT. The report presents the dataset, the main implementation design, and the evaluation setting used to compare these approaches.

Keywords: Computer Vision, Optical Character Recognition, Information Extraction, Receipt Understanding, Document Understanding.

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1 Introduction

Receipt understanding is a document analysis task in which the system must read a scanned or photographed receipt and recover specific pieces of information that can later be used in applications such as accounting, expense tracking, or business process automation.

In this project, we focus on extracting four key fields from each receipt: *company*, *date*, *address*, and *total*. Although the target output is simple, the task itself is not straightforward. Receipt images often contain small text, noisy backgrounds, irregular layouts, blur, shadows, and formatting differences across stores. These characteristics make the extraction stage much harder than plain OCR.

2 Dataset

2.1 Overview

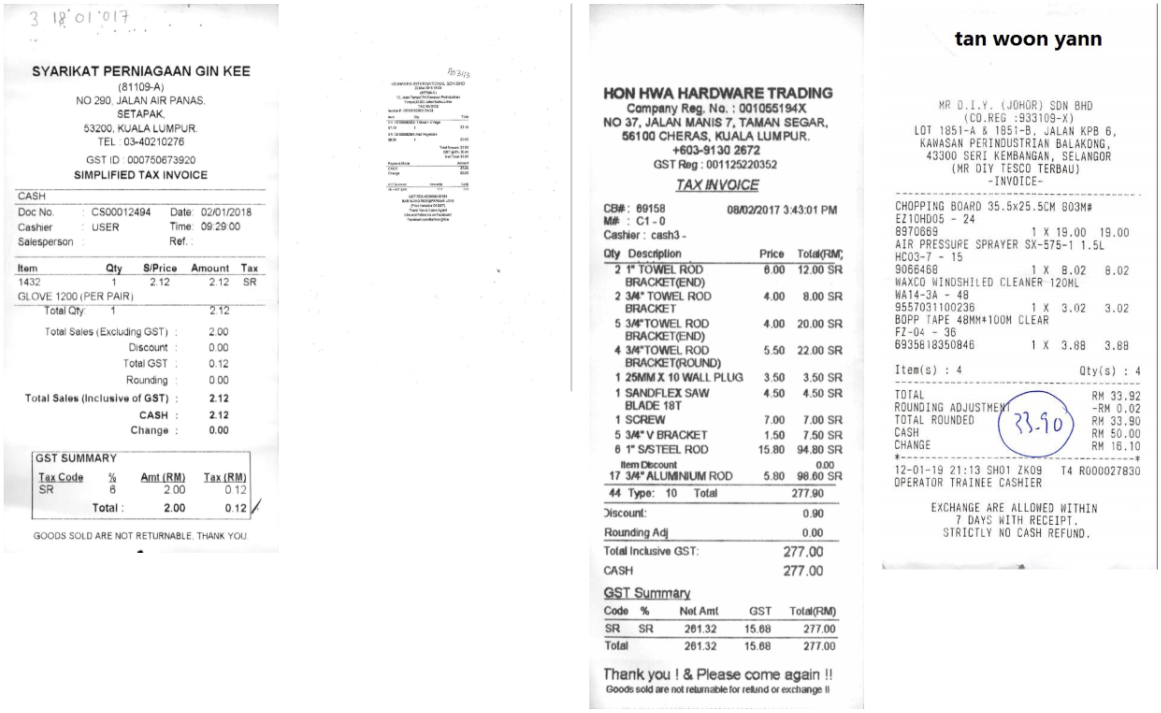

Figure 1: Sample receipt images from the SROIE dataset.

We use the SROIE2019 dataset, a benchmark dataset for scanned receipt OCR and information extraction. The dataset contains 973 receipt images in total, divided into 626 images for training and validation and 347 images for testing. The annotated text mainly consists of digits and English characters.

Each receipt image is accompanied by two annotation files. The first is a file containing bounding boxes together with the text associated with each box. The second is a JSON file containing the four target fields required for information extraction: **company**, **date**, **address**, and **total**.

The dataset is challenging for several reasons:

- The number of labeled images is still limited for training document understanding models.
- Each receipt contains many text boxes, often densely packed together.
- Text lines are usually narrow in height but highly variable in width.
- Image quality varies significantly across receipts, with blur, shadows, low contrast, and inconsistent formatting.

Preview Code Blame Executable File · 44 lines (44 loc) · 1.87 KB

```
148,48,564,48,564,86,148,86,HOME MASTER HARDWARE &
266,89,444,89,444,125,266,125,ELECTRICAL
85,121,625,121,625,156,85,156,NO.113G & 115G, JALAN SETIA GEMBILANG
166,161,540,161,540,194,166,194,U13/BG BANDAR SETIA ALAM,
175,204,530,204,530,235,175,235,40170 BANDAR SETIA ALAM,
275,240,428,240,428,267,275,267,SELANGOR.
151,282,553,282,553,315,151,315,COMPANY REG NO.:SA03711551-P
179,326,521,326,521,360,179,360,GST REG NO.:001637511168
263,382,433,382,433,409,263,409,TAX INVOICE
62,427,201,427,201,456,62,456,INVOICE NO.:
250,433,367,433,367,458,250,458,CS 67832
62,470,135,470,135,498,62,498,DATE :
248,475,453,475,453,500,248,500,22/12/2017 14:03
61,513,187,513,187,542,61,542,CASHIER#:
248,520,367,520,367,545,248,545,CASHIER
465,565,517,565,517,594,465,594,RM
562,563,636,563,636,593,562,593,CODE
58,606,341,606,341,636,58,636,24MMX7Y M.ONE TAPE
91,648,139,648,139,672,91,672,1.00
253,653,271,653,271,675,253,675,X
304,651,370,651,370,675,304,675,15.90
448,651,514,651,514,676,448,676,15.90
594,648,635,648,635,674,594,674,SR
181,696,295,696,295,722,181,722,SUBTOTAL :
448,698,512,698,512,722,448,722,15.90
85,743,294,743,294,772,85,772,TOTAL EXCL. OF GST
446,749,513,749,513,773,446,773,15.00
92,785,293,785,293,816,92,816,TOTAL INCL. OF GST
446,791,513,791,513,817,446,817,15.90
70,832,293,832,293,861,70,861,TOTAL AMT ROUNDED
444,838,509,838,509,860,444,860,15.90
173,874,293,874,293,906,173,906,PAYMENT :
443,876,511,876,511,901,443,901,50.00
132,911,291,911,291,945,132,945,CHANGE DUE :
442,915,510,915,510,940,442,940,34.10
60,959,237,959,237,995,60,995,TOTAL ITEM(S) : 1
61,1009,83,1009,83,1034,61,1034,G
140,1014,234,1014,234,1044,140,1044,UMMARY
268,1012,421,1012,421,1042,268,1042,AMOUNT(RM)
479,1010,589,1010,589,1043,479,1043,TAX (RM)
64,1057,103,1057,103,1081,64,1081,SR
155,1058,231,1058,231,1090,155,1090,@ 6%
355,1063,419,1063,419,1086,355,1086,15.00
534,1062,588,1062,588,1086,534,1086,0.90
```

Figure 2: Sample bounding boxes and corresponding OCR annotation file.

Code Blame Executable File · 6 lines (6 loc) · 191 Bytes

```
{
    "company": "LIGHTROOM GALLERY SDN BHD",
    "date": "20/11/2017",
    "address": "NO: 28, JALAN ASTANA 1C, BANDAR BUKIT RAJA, 41050 KLANG SELANGOR D.E, MALAYSIA",
    "total": "39.80"
}
```

Figure 3: Sample entity annotation containing the four target fields.

2.2 Exploratory Data Analysis (EDA)

2.2.1 Receipt Layout Characteristics

Receipts generally follow a semi-structured layout where key information appears in consistent regions. To capture these layout patterns, spatial features were derived from the bounding box coordinates. The center point and the line order of each bounding box were computed and normalized to the range [0,1] for consistence:

- **x_norm**: Normalized horizontal position
- **y_norm**: Normalized vertical position
- **line_norm**: Normalized line order within each receipt

These normalized layout features allow the system to identify regions of interest within the receipt variant and improve rule-based extraction.

To construct labeled training data for analysis, each OCR text box was assigned one of five labels: `company`, `address`, `date`, `total`, `other`. The labeling process was performed automatically by comparing the OCR text with the ground truth entities provided in the dataset. The labeling rules were defined as follows:

- If the OCR text contains the ground truth `total` value, the OCR box is labeled as **total**.
- If the OCR text matches a date pattern (e.g., `DD/MM/YYYY` or `DD-MM-YYYY`), it is labeled as **date**.
- If any word from the ground truth `company` name appears in the OCR text, the segment is labeled as **company**.
- If any word from the ground truth `address` appears in the OCR text, the segment is labeled as **address**.
- Otherwise, the OCR box is labeled as **other**.

This rule-based labeling strategy allows each OCR text segment to be associated with its corresponding entity type while preserving the spatial layout information. Table 1 illustrates the distribution for OCR Text Box Labels:

Table 1: Distribution of OCR Text Segment Labels

Label	**Number of Text Box**
Other	25682
Address	2475
Total	1542
Company	1395
Date	754
Total	31848

These labeled text segments are then used for exploratory analysis and for developing rule-based extraction methods that leverage both textual patterns and spatial layout information within receipts.

Figure 4 shows that the OCR boxes containing entities are typically located in the following areas:

- **Company** - Located at the top portion of the receipt: `line_norm` < 0.25.
- **Address** - Appears below the company name: `line_norm` < 0.35.
- **Date** - Usually found at the top or bottom section.
- **Total** - Typically located in the bottom half of the receipt: `line_norm` $>= 0.5$.

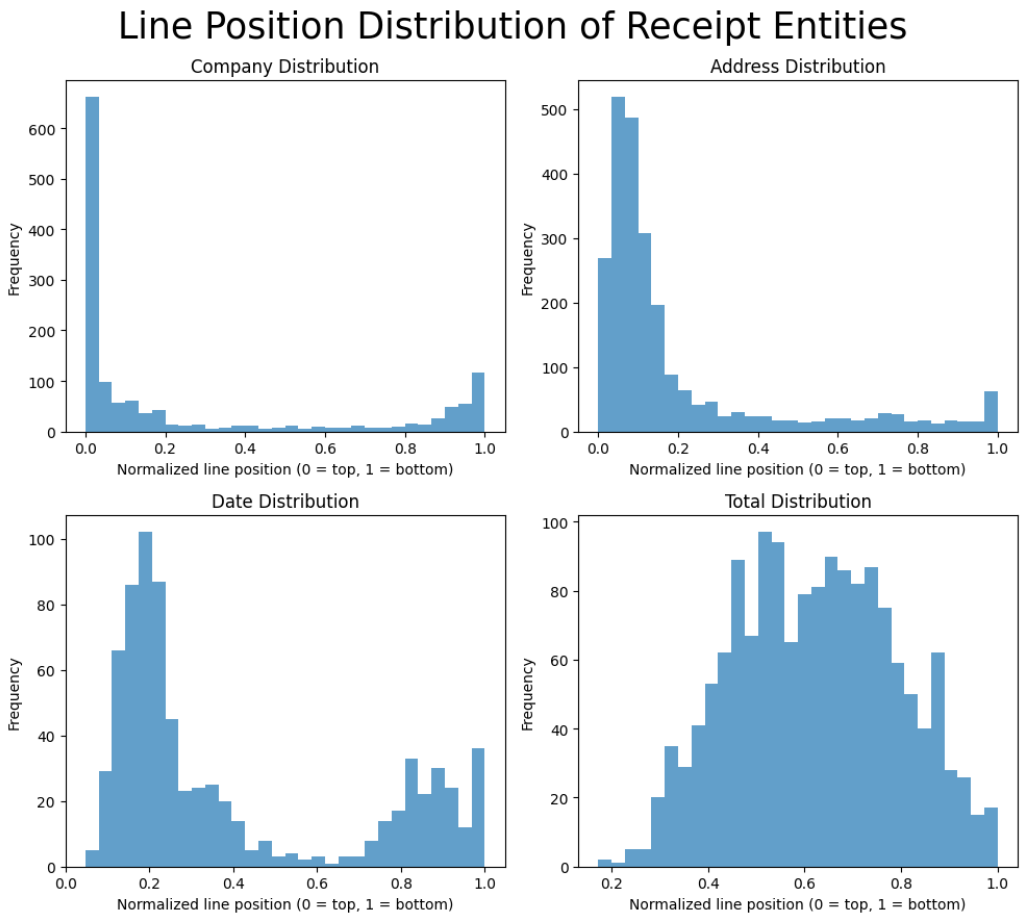

Figure 4: Normalized Line Position Distribution

2.2.2 Textual Characteristics

The OCR text extracted from receipts exhibits several patterns that can be used for entity identification.

1. **Pattern-Based Identification**

   Certain entities follow consistent textual patterns that can be detected using regular expressions. In particular, the **date** and **total** fields often exhibit distinctive numeric structures.

   Dates in receipts typically appear in formats such as:

   - DD/MM/YYYY
   - DD-MM-YYYY
   - DD/MM/YY

   These patterns can be captured using a regular expression such as:

   ```
   \d{2}[/-]\d{2}[/-]\d{2,4}
   ```

   Similarly, the **total amount** usually appears as a monetary value containing a decimal point. For example:

- 12.50
- 105.90
- 3.80

These values often follow the pattern:

```
\d+\.\d{2}
```

Detecting these numeric patterns helps identify candidate text segments that may correspond to the date or total entities.

2. **Keyword-Based Identification**

   Keywords also play an important role in identifying specific entities within the receipt text.

   For address fields, they frequently contain location-related words. These keywords often indicate that the text segment is part of the merchant's address. The most common keywords which is useful for rule in **address** entity are:

Table 2: Frequency of Address-Related Keywords

Keyword	Frequency
no	347
jalan	268
selangor	195
johor	116
alam	108
lot	107
lumpur	89
kuala	86

For the **company** entity, the OCR text is compared with the ground truth company name, and keyword matching is performed by checking whether any words from the company name appear in the OCR text segment. These are keywords that appear the most in the **company** entity and useful for defining rules:

Table 3: Frequency of Company-Related Keywords

Keyword	Frequency
sdn	305
bhd	356
gardenia	82
aeon	71
sanyu	58
stationery	47
co	45
restaurant	32

2.3 Data Quality Issues

Several data quality challenges were observed during exploratory analysis.

2.3.1 OCR Noise

OCR outputs sometimes contain spelling errors or character misrecognition. For example, the word `TOTAL` may be incorrectly recognized as `TOTAL`.

2.3.2 Encoding Issues

Some OCR files are encoded using `latin-1` rather than UTF-8, which may cause decoding errors when reading the files. To handle this issue, files were read using a more robust encoding strategy.

2.3.3 Irregular Formatting

Receipts vary significantly in layout, including differences in the number of lines, text alignment, and entity placement. This variability makes the extraction task challenging and motivates the use of layout-aware rules.

2.4 Summary

The exploratory data analysis revealed several important insights:

- Receipt layouts are semi-structured, with entities typically appearing in consistent regions.
- Dates and totals exhibit strong numeric patterns that can be detected using regular expressions.
- Address fields often contain location-related keywords.
- OCR outputs may contain noise and formatting inconsistencies.

These observations guided the design of the rule-based information extraction system developed in this project.

3 Methodology

In the current version of the project, the implementation is organized around three main directions for receipt understanding on the SROIE task. The first is a rule-based system used as a lightweight baseline at test time. The second is a layout-aware extraction model based on LayoutLMv3. The third is an end-to-end document understanding model based on DONUT. Although these branches process the input differently, they all target the same four fields: *company*, *date*, *address*, and *total*.

3.1 Implementation Overview

To make the structure of the project easier to follow, Table 4 summarizes the role of each main notebook in the current repository.

Table 4: Main implemented components in the current codebase

Notebook	**Main branch**	**Purpose**
`data_preprocessing.ipynb`	Data preparation	Reorganizes the original SROIE files into a cleaner structure so that each receipt image is consistently matched with its OCR box file and entity annotation file.
`ocr.ipynb`	OCR support pipeline	Applies image preprocessing and PaddleOCR to generate recognized text, bounding boxes, confidence scores, and OCR visualizations. This output is mainly used to support inspection and the rule-based baseline at test time.
`rule-based-system.ipynb`	Rule-based baseline applied at test time	Does EDA for the dataset to define the rule for the system. Then uses that rule to extract information from receipt images.
`layoutlm.ipynb`	Layout-aware extraction	Uses OCR tokens together with their bounding boxes to predict field labels with fine-tuned LayoutLMv3.
`DONUT.ipynb`	End-to-end extraction	Uses fine-tuned DONUT to learn a direct mapping from receipt image to a structured target sequence.

This organization reflects the way the project was developed. Instead of relying on a single pipeline, we explored the task through a simple baseline first, then through a layout-aware model, and finally through an end-to-end document understanding model.

3.2 Data Preparation

Before training or testing any model, the original SROIE files are first reorganized into a more convenient structure. Each receipt is associated with three aligned files: the image itself, the OCR box annotation file, and the entity annotation file. The preprocessing notebook standardizes file naming and separates these files into dedicated folders so that later steps can load the correct inputs and labels without manual matching.

This stage is important because the rest of the pipeline depends on consistent alignment between images and annotations. If one receipt image is paired with the wrong OCR file or the wrong entity file, the later extraction results become meaningless even if the model itself is working properly.

3.3 Rule-based System

We place the rule-based system first because it is the simplest baseline in the project. Unlike the learning-based branches, this system is not designed as a full training pipeline. It is applied only at test time and serves as a straightforward reference point for comparison.

The rule relies on OCR output and the EDA for the dataset. In practice, the input text is obtained from the OCR support pipeline, then checked with the rule extracted from EDA to see which one it belong in the four target fields. This design keeps the system simple and interpretable, although it also makes it sensitive to OCR errors and formatting variation.

3.3.1 OCR Support for the Rule-based Branch and the EDA

Before rules can be applied, the receipt image is passed through a lightweight OCR preparation pipeline. The image is preprocessed with OpenCV to make text regions easier to read, and PaddleOCR is then used to detect and recognize text.

The preprocessing operations used in this supporting step are summarized in Table 5.

Table 5: Image preprocessing operations used before OCR

Operation	**Purpose**
Grayscale conversion	Simplifies the image by removing color variation and keeping only intensity information.
Gaussian blur	Reduces small-scale noise and local artifacts that may interfere with OCR.
CLAHE contrast enhancement	Improves local contrast so that faint or unevenly illuminated text becomes easier to read.

After preprocessing, PaddleOCR returns the recognized text together with bounding boxes and confidence scores. The raw OCR output is then cleaned before rule application. The main post-processing steps are shown in Table 6.

Table 6: OCR post-processing steps for the rule-based branch

Step	**Role in the pipeline**
Sorting OCR lines	Reorders text lines from top to bottom and left to right so that the output follows a natural reading order.
Filtering low-confidence predictions	Removes OCR outputs that are too uncertain and likely to introduce noise.
Basic cleaning	Discards very short fragments and trims unnecessary spaces.
Structured export	Saves cleaned OCR results in a more usable format for later inspection and rule application.

At the end, EDA is performed by the OCR output of the training set and the entities from the original training dataset.

3.3.2 Rule Definition and Information Extraction

Once OCR text for test dataset has been cleaned, the system directly applies hand-crafted rules to extract the four target fields. These rules are based on simple textual patterns and heuristic assumptions about where the information usually appears.

For example, the system can search for common date patterns, look for total-like numeric strings like 12.50 or 105.90, or select the company name from the upper part of the receipt. This branch is easy to interpret because each prediction can be traced back to a specific rule. At the same time, its accuracy depends strongly on OCR quality and on how closely the receipt follows common layout conventions.

3.4 LayoutLMv3-based Extraction

The second branch utilizes LayoutLMv3, a multi-modal model that jointly learns from text, layout, and image features. This approach is particularly well-suited for receipt information extraction, where the semantic meaning of a field is often defined not just by its textual content but also by its spatial position (e.g., total amounts usually appear at the bottom right).

In our implementation, the pipeline consists of three main stages: data preprocessing and labeling, model fine-tuning, and post-processing for field reconstruction.

3.4.1 Data Preprocessing and Labeling

Since the raw SROIE dataset provides annotations at the line level, while LayoutLMv3 operates at the token level, several preprocessing steps are required to align the inputs:

- **Text Normalization:** To reduce vocabulary noise, we apply a normalization function that converts all text to lowercase and retains only alphanumeric characters (‘[a-z0-9]‘). This helps the model focus on semantic content rather than formatting variations.
- **Word-Level Splitting:** The original OCR lines are split into individual words using whitespace as a delimiter. To maintain spatial alignment, the bounding box of the original line is duplicated for each word it contains. While this is an approximation, it effectively provides the model with the general spatial region of each token.
- **Box Normalization:** Bounding boxes are normalized to a $[0, 1000]$ scale relative to the image width and height, consistent with the LayoutLMv3 pre-training requirements.
- **BIO Tagging Scheme:** We utilize the BIO (Beginning, Inside, Outside) tagging scheme to transform entity values into token-level labels. The entity annotation file is loaded, and for each field (Company, Address, Date, Total), we locate its corresponding tokens in the document. The first token of an entity is labeled with a ‘B-‘ tag (e.g., ‘B-COMPANY‘), subsequent tokens with ‘I-‘ (e.g., ‘I-COMPANY‘), and irrelevant tokens are labeled ‘O‘.

3.4.2 Input Construction

The OCR annotation file is first loaded to obtain text lines and their boxes. These lines are then split into words, and the corresponding bounding boxes are duplicated so that each word has its own aligned box. The entity annotation file is loaded separately and used to create token-level labels.

To make this design easier to read, the main input structure for this branch is summarized in Table 7.

Table 7: Input design in the LayoutLMv3 branch

Component	**Description**
Receipt image	Original document image used by the model.
OCR words	Word-level tokens derived from the OCR annotation file.
Bounding boxes	Spatial coordinates aligned with each token and normalized for the processor.
Field labels	Token-level supervision created from the entity annotations.
Model output	Predicted token labels that are later merged into the four final fields.

3.4.3 Fine-tuning Configuration

We fine-tune the pre-trained ‘microsoft/layoutlmv3-base‘ model using the Hugging Face Transformers library. The dataset is split into training and validation sets with a 90/10 ratio. The specific hyperparameters used for training are detailed in Table 8.

Table 8: Hyperparameters for LayoutLMv3 Fine-tuning

Parameter	**Value**
Base Model	`microsoft/layoutlmv3-base`
Learning Rate	3×10^{-5}
Batch Size (per device)	2
Num Epochs	10
Max Sequence Length	512
Optimizer	AdamW
Scheduler	Linear Decay

3.4.4 Prediction Reconstruction

During inference, the model predicts a label for each token in the input sequence. We reconstruct the final entity values by aggregating tokens based on their predicted tags:

1. **Token Aggregation:** We iterate through the predicted tags. When a ‘B-‘ tag is encountered, a new entity candidate is initialized. Subsequent tokens labeled with the matching ‘I-‘ tag are appended to this candidate.

2. **Merging:** The collected tokens for each entity are joined by spaces to form the final string value.

3. **Filtering:** Any entities that do not follow valid BIO transitions (e.g., an ‘I-‘ tag appearing without a preceding ‘B-‘) are discarded or handled via heuristic rules to ensure output consistency.

3.5 DONUT-based End-to-End Extraction

The third branch uses a fine-tuned DONUT model, which is an end-to-end document understanding model. This branch is placed last because it is the most self-contained one: it does not depend on OCR during inference. Instead, it receives the receipt image directly and learns to generate a structured output sequence containing the target fields.

3.5.1 Data Preprocessing and Special Tokens

Unlike traditional OCR pipelines, DONUT utilizes a specific preprocessing pipeline handled by the `DonutProcessor`. We perform the following steps to prepare the data:

- **Image Resizing:** Input images are converted to RGB and resized to a fixed resolution of 1280×960. We enable `do_align_long_axis` to ensure the aspect ratio is preserved during resizing, which is critical for document readability.
- **Special Tokens:** We extend the tokenizer's vocabulary with task-specific tokens (e.g., `<s_company>`, `</s_company>`, etc.). The model's embedding layer is resized to accommodate these new tokens, allowing it to treat field delimiters as atomic units.
- **Label Encoding:** Target sequences are tokenized, and padding tokens are masked with indices of -100 so they are ignored during the cross-entropy loss calculation.

3.5.2 Target Construction

Each training sample consists of an image path and a target string. The target string wraps the four fields inside special tags, enclosed within a root receipt tag. The format is summarized in Table 9.

Table 9: Target format used in DONUT training

Field	Tagged format
Company	`<s_company> ... </s_company>`
Date	`<s_date> ... </s_date>`
Address	`<s_address> ... </s_address>`
Total	`<s_total> ... </s_total>`
Full output	`<s_receipt> ... </s_receipt>`

3.5.3 Fine-tuning Configuration

We initialize the model using the pre-trained `naver-clova-ix/donut-base` checkpoint. The model is fine-tuned using the `Seq2SeqTrainer` with the hyperparameters detailed in Table 10.

Table 10: Hyperparameters for DONUT Fine-tuning

Parameter	Value
Base Model	`naver-clova-ix/donut-base`
Batch Size	10
Learning Rate	3×10^{-5}
Epochs	10
Precision	FP16 (Mixed Precision)
Optimizer	AdamW

3.5.4 End-to-End Generation

This design turns the extraction problem into a sequence generation problem. Instead of labeling each token separately, the model learns to produce the final structured answer directly. This makes the DONUT branch the most independent of the three, since it learns the mapping from receipt image to structured output without a separate OCR stage at inference time.

3.6 Output Normalization

Before evaluation, both predictions and ground-truth values are normalized. In the current implementation, this mainly includes converting text to lowercase, trimming leading and trailing spaces, and collapsing repeated whitespace. These steps are simple, but they are necessary because receipt annotations often differ in casing or spacing even when the actual content is the same.

Without normalization, the evaluation would become too strict and would penalize differences that are mostly cosmetic rather than meaningful.

3.7 Evaluation Metrics

The models are evaluated at field level after normalization. Since the report later compares different branches, it is useful to present the metrics clearly from the beginning. Table 11 summarizes the main metrics used in the current implementation.

Table 11: Evaluation metrics used in the current implementation

Metric	Meaning	Level
Field Exact Accuracy	Measures how often a predicted field matches the normalized ground truth exactly.	Per field
Precision	Measures how many predicted entities are correct.	Overall extraction
Recall	Measures how many ground-truth entities are recovered.	Overall extraction
F1 Score	Balances precision and recall into a single score.	Overall extraction
Normalized Levenshtein Distance	Measures how different a prediction is from the ground truth even when exact match fails.	String distance

3.7.1 Field Exact Accuracy

For each field $f \in \{\text{company}, \text{date}, \text{address}, \text{total}\}$, field accuracy is computed as

$$\text{Accuracy}_f = \frac{\text{Number of correct predictions for } f}{\text{Total number of evaluated samples}}.$$

This is the strictest metric in the report. A field is counted as correct only when the predicted value is exactly the same as the normalized reference value. This is appropriate for receipt understanding because even a small mismatch may already make the field unusable in practice.

3.7.2 Precision, Recall, and F1

Besides field-wise accuracy, the current implementation also reports precision, recall, and F1 in order to give a broader view of extraction quality. If C denotes the number of correctly predicted entities, P the number of predicted entities, and G the number of ground-truth entities, then

$$\text{Precision} = \frac{C}{P}, \qquad \text{Recall} = \frac{C}{G}, \qquad \text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

These metrics are useful because they reveal whether the model is too conservative or too noisy overall, which exact match alone does not always show.

3.7.3 Normalized Levenshtein Distance

Exact match is often too harsh for long fields such as addresses. For that reason, the evaluation also includes a normalized Levenshtein distance. Given a predicted string p and a ground-truth string g, the distance is defined as

$$\text{LevDist}(p,g) = \frac{\text{LevenshteinDistance}(p,g)}{\max(|p|,|g|)}.$$

The final score is averaged across the evaluated fields. A value closer to 0 indicates that the prediction is closer to the reference, while a higher value indicates a larger difference between the two strings.

3.8 Evaluation Protocol

Although the three branches are different internally, the evaluation logic is kept consistent:

1. load the receipt sample and its ground-truth entities,
2. run one extraction branch,
3. normalize both the prediction and the reference,
4. compare the four target fields using the metrics above.

This makes the comparison between the rule-based system, LayoutLMv3, and DONUT more fair, because the same output format and the same evaluation rules are applied throughout the project.

4 Results

4.1 Field Accuracy

Table 12: Test Set Evaluation: Field Accuracy

Model	company ↑	date ↑	address ↑	total ↑
LayoutLMv3	0.285	0.135	0.037	0.669
Rule-based Extraction	0.542	0.156	0.009	0.418
DONUT	0.775	0.816	0.550	0.703

- Table 12 reports field-level accuracy for the four target fields: company, date, address, and total.
- Among the four fields, total appears to be easier to identify than the others, while address remains the most difficult one because of its variable length and inconsistent formatting across receipts.
- The LayoutLMv3 model performs poorly overall, especially on the address field, where the accuracy is extremely low. Its performance on the company field is also limited, which suggests that variations in store names and receipt layout still make this field difficult for the model.
- Rule-based extraction performs relatively well on the company field, but its performance on the date and address fields is weak. Its result on the total field is also lower than those of the other two methods.
- DONUT performs relatively well across all four fields. Although the address field is still the weakest one, with an accuracy of 0.550, the model clearly outperforms the other two approaches on most fields.

4.2 Performance Analysis on Entity Metrics and Normalized Levenshtein Distance

Table 13: Entity Metrics and Normalized Levenshtein Distance

Model	Precision ↑	Recall ↑	F1 Score ↑	Normalized Levenshtein Distance ↓
LayoutLMv3	0.403	0.282	0.331	0.439
Rule-based Extraction	0.281	0.281	0.281	0.416
DONUT	0.711	0.711	0.711	0.102

- Table 13 presents four evaluation metrics for our three models: Precision, Recall, F1 Score, and Normalized Levenshtein Distance.
- The LayoutLMv3 model achieves a precision score of 0.403 and a recall score of 0.282. These results indicate that the model correctly identifies only a limited portion of the target fields, while still producing many incorrect or missing predictions. Its normalized Levenshtein distance of 0.439 shows that the extracted text still differs noticeably from the ground truth.

- The rule-based method is more consistent, with both precision and recall equal to 0.281. Although its F1 score remains low, this method achieves a slightly lower normalized Levenshtein distance of 0.416 than LayoutLMv3. This suggests that, even when the extracted fields are not exactly correct, their predicted values are on average slightly closer to the ground truth.
- DONUT produces the strongest overall result. Its precision, recall, and F1 score are all equal to 0.711, showing a stable extraction performance across the dataset. In addition, DONUT achieves the lowest normalized Levenshtein distance, 0.102, which indicates that its predicted outputs are much closer to the ground truth than those of the other two models. This is the most promising result in our current experiments.

5 Discussion and Future Works

5.1 Discussion

Despite the promising results of the DONUT (Document Understanding Transformer) model, several points for improvement remain:

- The performance of the two approaches that rely on OCR-based textual inputs is still relatively disappointing. We observe that PaddleOCR often returns reasonable text boxes and text content that are already quite close to the ground truth. This suggests that the later extraction stage, especially post-processing and field reconstruction, may not yet be sufficiently optimized for the task.
- Due to time and resource constraints, we were not able to train and tune all models as thoroughly as desired. Future work should therefore include more complete training and validation in order to obtain more reliable performance.
- We did not include a data augmentation step, assuming that the pre-trained models had already benefited from sufficient augmentation during pre-training. However, the current results suggest that task-specific data augmentation may still be important for improving performance, even when fine-tuning pre-trained models.

5.2 Future Works

5.2.1 Further Experiments on the Training Process

We were able to experiment with three models for this task. One model, DONUT, directly generates the result from the receipt image, while the other two approaches rely on OCR-based textual inputs before the final extraction step. In future work, we would like to explore additional fine-tuning strategies, such as continued fine-tuning or instruction tuning, since different users may require different types of information from receipts.

5.2.2 Practical Application

Although this task is informative and interesting from a research perspective, its immediate practical use is still limited for several reasons:

- Most receipts in the dataset are in English, so the system is not directly applicable to Vietnamese receipts.
- The task is also quite research-oriented, since it extracts only four fields: company, date, address, and total. This limited information is not sufficient for many realistic use cases, such as maintaining a complete receipt database for a company or tracking a person's financial activities in detail.

In the future, our team aims to make the system more practical for real-world use. One possible direction is to build a dataset specifically for Vietnamese receipts and generate corresponding annotations for downstream tasks. For example, the system could support personal finance management by classifying each expense type and combining extraction outputs with a large language model to provide users with advice on improving their spending habits.

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